2022(t), 2022
(X) Explain sensitized fluorescence and quenching of fluorescence ?

⇒ A substance which is normally non-fluorescent may be made fluorescent ~~inth~~ in the presence of other fluorescent substances. The phenomenon is known as sensitized fluorescence. If the vapour of thallium is added to mercury vapour and then exposed to radiations of wave-length 2537Å, the vapour of thallium fluoresces. Thus the thallium which is a non-fluorescent substance can be made ~~by~~ fluorescent by mercury vapour. Mercury vapour gets energy from incident light and thus its atoms get excited to higher energy levels.

$$Hg(^1S_0) + h\nu \longrightarrow Hg^*(P_1)$$

thallium atoms undergo collision with excited mercury

atoms and thus a part of energy of excited mercury atoms is transferred to thallium atoms which are raised to higher energy levels.

$$Hg^* + Tl \longrightarrow Tl^* + Hg$$

These excited atoms of thallium emit their own characteristics fluorescence on reverting to the ground state.

$$Tl^* \longrightarrow Tl + h\nu$$

⇒ when a photochemical excited atom has a change to undergo collision with another atom or a molecule before it fluoresces, the intensity of the fluorescent radiation may be diminished or stopped. this pheiome is known as quencing of fluorescence. some ~~fact~~ facts about quenching of f. are as follows!

① quenching of f. depends greatly on the concentration of the fluorescent atoms and of the quenching substance.

② At low pressures, very little quenching will occur.

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At high pressures, appreciable quenching takes place.

(iii) Q. of f. takes place appreciably in a liquid medium because collisions are very frequent.

2021(t), 2021

(*) The dissociation energy of hydrogen is 102.9 Kcal/mol. If H_2 is dissociated by illumination with radiation of $\lambda = 2537\,Å$. what fraction of the radiant energy will be converted into K.E.?

⇒ Dissociation energy per molecule $= \frac{102.9 \times 10^3}{6.023 \times 10^{23}}$ cal

$= 1.7 \times 10^{-19} \times 4.2$ J

$= 7.112 \times 10^{-19}$ J

The energy of one quanta, $E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{2537 \times 10^{-10}}$ J

$= 7.84 \times 10^{-19}$ J

∴ Energy converted into K.E. $= (7.84 \times 10^{-19} - 7.112 \times 10^{-19})$ J

$= 7.232 \times 10^{-20}$ J

∴ fraction converted into K.E $= \frac{7.232 \times 10^{-20}}{7.84 \times 10^{-19}}$

$= 0.092$ Ans.

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(★) 2022, 2020
The dissociation energy of a molecule is 142.95 Kcal mol^{-1}. What is the longest λ of light capable of dissociation of this molecule:

$\Rightarrow E = 142.95 \text{ Kcal mol}^{-1} = \frac{142.95 \times 1000 \times 4.2}{6.023 \times 10^{23}} \text{ J}$

$= 9.97 \times 10^{-19} \text{ J}$

Now, $E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$

$= 1.994 \times 10^{-7} \text{ m}$ Ans.

(★) 2017, 2019(t)
The quantum yield of the reaction, $H_2(g) + Br_2 \rightarrow 2HBr$ is 1×10^6 with $\lambda = 480 \text{ nm}$. calculate the number of moles of HBr(g) produced per Joule of radiant energy absorbed

$\Rightarrow$ Energy absorbed per mole $= \frac{Nhc}{\lambda}$

$= \frac{6.023 \times 10^{23} \times 6.626 \times 10^{-34} \times 3 \times 10^8}{480 \times 10^{-9}}$

$= 2.49 \times 10^5 \text{ J mol}^{-1}$

$Q = \frac{\text{No. of moles reacted}}{\text{No. of einsteins absorbed}}$

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⇒ No. of mole reacted = Q × No. of einsteins absorbed

Moles of HBr produced = 2 × no. of mole of H_2 or Cl_2 reacted

$= 2 \times Q \times$ No. of einsteins absorbed

$= 2 \times 1 \times 10^6 \times \frac{1}{2.49 \times 10^5}$

$= 8.03$ moles Ans.

2022, 2020, 2017, 2015, 2019(t), 2022(t)

(A) Discuss Mechanism and deduce kinetics of photochemical reaction btwn H_2 and Br_2 ?

⇒ Mechanism:

	Rate constant	Rate law
1. Initiation:		
(i) $Br_2 + h\nu \longrightarrow 2Br$	K_1	Rate $= K_1 I_{abs}$
2. Propagation:		
(ii) $Br + H_2 \longrightarrow HBr + H$	K_2	Rate $= K_2[Br][H_2]$
(iii) $H + Br_2 \longrightarrow Br + H$	K_3	Rate $= K_3[H][Br_2]$
3. Inhibition:		
(iv) $H + HBr \longrightarrow Br + H_2$	K_4	Rate $= K_4[H][HBr]$
4. Termination:		
(v) $Br + Br \longrightarrow Br_2$	K_5	Rate $= K_5[Br]^2$

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⇒ Deduce Rate Law:

$$\frac{d[HBr]}{dt} = K_2[H_2][Br] + K_3[H][Br_2] - K_4[H][HBr] \quad \text{---} (1)$$

under steady-state condition

$$\frac{d[H]}{dt} = K_2[Br][H_2] - K_3[H][Br_2] + K_4[H][HBr]$$

$$K_4[H][HBr] = 0 \quad \text{---} (2)$$

and,

$$\frac{d[Br]}{dt} = K_1 I_{abs} - K_2[Br][H_2] + K_3[H][Br_2] + K_4[H][HBr] - K_5[Br]^2 = 0 \quad \text{---} (3)$$

adding (2) & (3), we get,

$$K_1 I_{abs} - K_5[Br]^2 = 0$$

$$\therefore [Br] = \left(\frac{K_1}{K_5} \cdot I_{abs}\right)^{1/2} \quad \text{---} (4)$$

From eqn (2), we get, $[H] = \dfrac{K_2[Br][H_2]}{K_3[Br_2] + K_4[HBr]} \quad \text{---} (5)$

putting value of [Br] from (4) in (5), we get,

$$[H] = \frac{K_2\left(\frac{K_1}{K_5} \cdot I_{abs}\right)^{1/2}[H_2]}{K_3[Br_2] + K_4[HBr]} \quad \text{---} (6)$$

From eqn (1), (4) & (6), we get,

$$\frac{d[HBr]}{dt} = -\frac{2K_2\sqrt{K_1 \cdot I_{abs}/K_5} \cdot [H_2]}{1 + (K_4[HBr]/K_3[Br_2])}$$

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This is the rate-law eqn for overall reaction..

⭐ 2020, 2016
Differentiate btwn photochemical & thermal reaction?

Photochemical reaction	Thermal reaction
The reaction involve absorption of light radiations.	The reactions involve absorption or evolution of heat energy.
presence of light is the primary requirement.	These reaction can take place in light as well as in dark.
Rate of reaction is independent on temperature	Rate of reaction depends on temperature.
ΔG may be +ve or -ve	ΔG is always -ve.
photochemical activation is highly selective process	Thermal activation is not selective in nature.
Require light as an external energy source	Depend on heat energy from an external source.
Depend on light λ and intensity	Depend on T and heat capacity.
Initiated by light absorption	Initiated by thermal energy input
often produce specific products due to selective excitation	May produce a range of products depending on the conditions.
example: photosynthesis, photodecomposition of silver halides	combustion; thermal decomposition of $CaCO_3$

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(*) 2021, 2018, 2021(t), 2016, 2018(t)
Deduce expression for the kinetics of photochemical btwn Cl_2 & H_2 in absence of oxygen?

⇒ Mechanism of Reaction:

Case 1: In absence of oxygen

(a) primary process (initiation):

	Rate constant	Rate law
(i) $Cl_2 + h\nu \longrightarrow 2Cl$	K_1	$Rate = K_1 I_{abs}$

(b) secondary process:

1. chain propagation

(ii) $Cl + H_2 \longrightarrow HCl + H$	K_2	$Rate = K_2[Cl][H_2]$
(iii) $H + Cl_2 \longrightarrow HCl + Cl$	K_3	$Rate = K_3[Cl_2][H]$

2. chain termination:

(iv) $Cl + wall \longrightarrow \frac{1}{2}Cl_2$	K_4	$Rate = K_4[Cl]$

The above mechanism is interpreted with the help of steady-state principle.

∴ Net rate of formation of Cl = (Rate of formation of Cl) − (Rate of removal of Cl)

$$\frac{d[Cl]}{dt} = K_1 I_{abs} + K_3[H][Cl_2] - K_2[Cl][H_2] - K_4[Cl]$$

But, at steady state: $\frac{d[Cl]}{dt} = 0$

$$\therefore K_1 I_{abs} + K_3[H][Cl_2] - K_2[Cl][H_2] - K_4[Cl] = 0 \quad ---(1)$$

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Similarly, $\frac{d[H]}{dt} = K_2[Cl][H_2] - K_3[H][Cl_2] = 0$ ---- (2)

Adding eqn (1) & (2) we get,

$$K_1 I_{abs} - K_4[Cl] = 0$$

$$[Cl] = \frac{K_1}{K_4} \cdot I_{abs} \quad ---- (3)$$

From eqn (2), $K_2[Cl][H_2] = K_3[H][Cl_2]$ ---- (4)

Net rate of formation of HCl is given by,

$$\frac{d[HCl]}{dt} = K_2[Cl][H_2] + K_3[H][Cl_2] \quad ---- (5)$$

From (4) & (5), $\frac{d[HCl]}{dt} = 2K_2[Cl][H_2]$ ---- (6)

From (3) & (6), $\frac{d[HCl]}{dt} = 2 \cdot \left(\frac{K_1 \cdot K_2}{K_4}\right) \cdot I_{abs}[H_2]$ ---- (7)

eqn (7) is the rate law of the overall reaction.

(*) 2022, 2020, 2022(t), 2021

Define photosensitized reactions & give examples using different types of photosensitizer?

⇒ The reactions which can be made light sensitive by adding a small amount of foreign material which can

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absorb light and stimulate the reaction without itself taking part in the reaction, is called photosensitize reactions. such an added material is known as photosensitizer and the phenomenon is photosensitization.

or, net,

P. reaction is a type of photochemical process in which a molecule, called photosensitizer, absorbs light energy and then transfer energy to another molecule, causing a chemical reaction. examples are:

⇒ (i) Mercury as photosensitizer:

When mixture of mercury vapour and H gas by light wavelength 253.7 nm the dissociation of molecular H into atomic H takes place.

$$Hg(g) + H_2 \longrightarrow 2H + Hg(g) \quad \text{at } 253.7\ nm$$

Hg atom absorbs light → Hg* collide with H_2 → dissociate into atoms.

Mechanism:

$$Hg + h\nu \longrightarrow Hg^*$$

$$Hg^* + H_2 \longrightarrow H_2^* + Hg$$

$$H_2^* \longrightarrow 2H$$

(ii) chlorine as photosensitizer: Decomposition of ozone

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in presence of chlorine in UV.

$$2O_3 + h\nu \xrightarrow{Cl_2} 3O_2$$

Rate of reaction independent of concentration but proportional to intensity of light.

Mechanism: $Cl_2 + h\nu \longrightarrow 2Cl$

$$Cl + O_3 + M \longrightarrow 2ClO_3^* + M$$

Excited ClO_3^* may absorbed on the wall to form Cl_2O_6, Cl_2 and O_2.

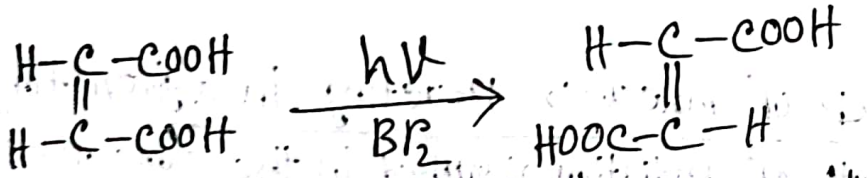
$$ClO_3^* + ClO_3^* + M \longrightarrow Cl_2 + 3O_2 + M$$

ClO_3^* may react to form ClO_2 and O_2

$$ClO_3^* + O_3 \longrightarrow ClO_2 + 2O_2$$

$$ClO_2 + O_3 \longrightarrow ClO_3 + O_2$$

(iii) Bromine as a photosensitizer (P.): Bromine act as a P. in the conversion of maleic acid into fumaric acid.



Maleic acid — Fumaric acid

(iv) Cadmium vapour as a P.: Cd vapour act as a P. for the polymerisation of ethylene.

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$$n\,C_2H_4 \xrightarrow[Cd]{h\nu} (-C_2H_2-)_n$$

(v) uranyl ion as a p.: uranyl ion act as a p. in the photolysis of formic acid.

$$UO_2^{2+} + h\nu \longrightarrow [UO_2^{2+}]^*$$

$$[UO_2^{2+}]^* + HOOC-COOH \longrightarrow CO_2 + CO + H_2O + UO_2^{2+}$$

(vi) chlorophyll as a p.: It act as a p. in the photosynthesis of of carbohydrate from CO_2 & H_2O.

$$\text{chlorophyll} + h\nu \longrightarrow [\text{chlorophyll}]^*$$

$$6CO_2 + H_2O + [\text{chlorophyll}]^* \longrightarrow C_6H_{12}O_6 + O_2 + \text{Chlorophyll}$$

(vii) 2021, 2021(t)
define quantum yield? which factor responsible for low & high quantum yield.?

⇒ Efficiency of a photochemical process is expressed interm of quantum efficiency or quantum yield (φ). It states that "The number of molecules reacting pe quantum of light absorbed."

$$\varphi = \frac{\text{Number of molecules reacting in a given time}}{\text{Number of quanta of radiation absorbed in the same time}}$$

$$= \frac{\text{Rate of chemical reactions}}{\text{'Einsteins' absorbed}}$$

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where 'Einsteins' is the unit of photochemical intensity. The value of φ may vary from 1 to 10^{10}; In some cases it is less than 1.

Reason for low φ:

(I) Excited molecules may get deactivated before they form a product so the value is less than 1.

(II) collision of excited molecule with non excited molecule.

(III) Molecules may not receive adequate energy to enable them to react

(IV) The photochemical reaction may be reversed.

Reason for High φ:

(I) After the absorption of radiation, the activated molecule which is formed may collide with another active molecule

(II) Due to the formation of intermediate product which acts as a catalyst.

(III) The radiation absorption includes the production of the atoms with initiate the chain reaction.

(IV) The product of primary process may collide with

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2nd molecule and transfer energy. 2nd to 3rd and so on. Thus, the chain reaction starts and number of reacting molecule will be high so φ will high.

(V) one photon absorbed in a primary reaction dissociate one molecule of the reactant. But the excited atoms may start a subsequent secondary reaction in which a further molecule is decomposed.

$$AB + h\nu \longrightarrow A + B \qquad \text{primary}$$

$$AB + A \longrightarrow A_2 + B \qquad \text{secondary}$$

(VI) when there are two or more reactants, a molecule of one of them absorbs a photon and dissociates. The excited atom that is produced starts a secondary reaction chain.

$$A_2 + h\nu \longrightarrow 2A \qquad \text{primary}$$

$$A + B_2 \longrightarrow AB + B \qquad \text{secondary}$$

$$B + A_2 \longrightarrow AB + A \qquad \text{Reaction chain}$$

2022, 2020, 2015, 2022(t)

⊛ Discuss the primary photochemical process using molecular spectra?

⇒ It is the process in which light radiation is

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absorbed by an atom or molecule giving rise to the formation of an excited atom or an excited molecule.

$$A + h\nu \rightarrow A^*$$

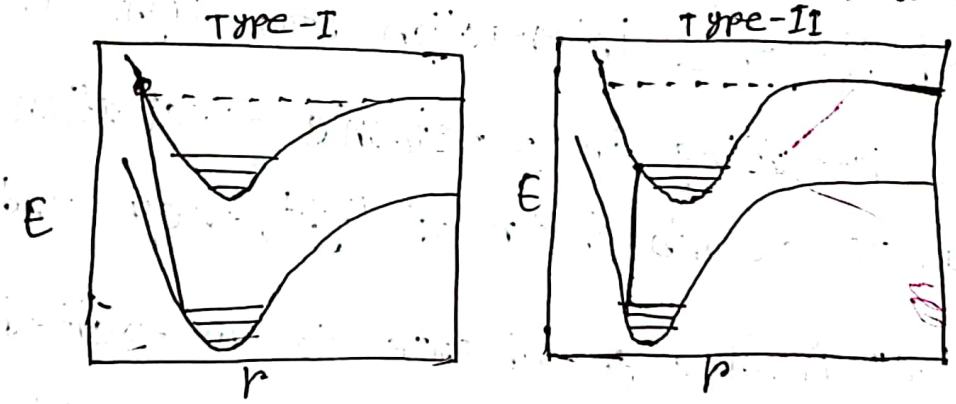
For photochemical reaction an electronic transition occurs in the visible or UV region. The nature of the resulting primary process the depends on the relationship btwn the upper & lower electronic states of molecule. The primary P. process is initiated by the interaction btwn light and molecular species, resulting in electronic excitation. when molecules absorb photon of light, they undergo electronic transitions from a ground state to an excited state. The energy of the absorbed photon must match the energy dfrnce btwn the ground state and an excited state which can be described by the molecular absorption spectrum. The transition often involves vibrational and rotational excitation as well. A Primary P. Process is the initial step in a photochemical reaction; where a molecule absorbs a photon of light and transitions to an

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electronically excited state. The primary P. reaction are e⁻ transfer reactions that creat oxidized chlorophylls and reduced acceptors. Four types of behaviour are of interested in connection with the primary P. process.

Type-I: In type-I, the nature of the potential energy curves is such that in the transition indicated the vibrational energy of the molecule in the upper electronic state exceeds the maximum value.



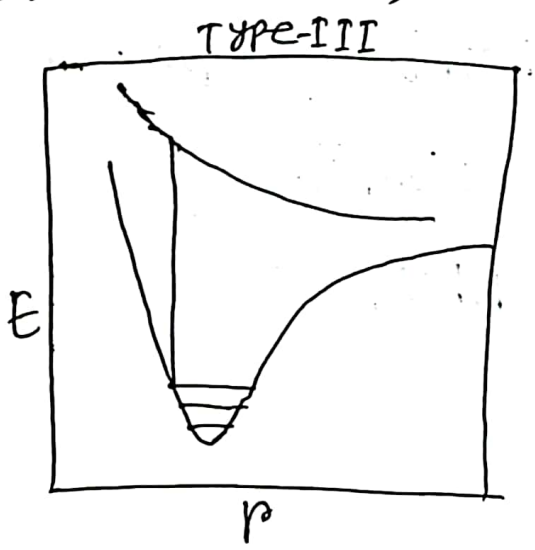
Type-II: In type-II, no transition from lower to upper state can result in dissociation of the molecule.

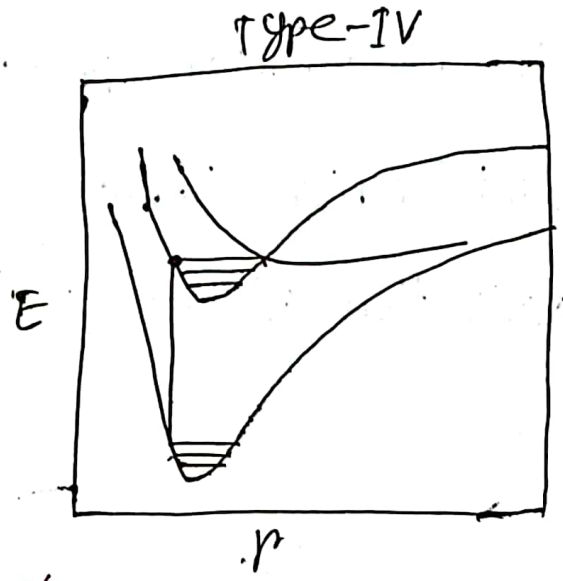
Type-III: Type-III is uncommon, the upper electronic state is completely unstable and has no vibrational level

Type-IV: It is the combination of type-II and type-I there are two upper electronic levels close together; one is like that in case II (stable) and other like

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case III (unstable).





2021, 2018, 2017, 2016, 2015, 2018(t)

Q. State the Grotthus-Draper law and the Einstein's law of photochemistry.? or State law of photochemistry?

⇒ Grotthus-Draper law: "only those radiation which are absorbed by a reacting substance or system are responsible for producing chemical change".

A. to this law, all light radiations are not bringing the chemical reaction. some are increase the kinetic energy of molecule while some are re-emitted (fluoresce

⇒ The law of photochemical equivalence states that: "whe an atom or molecule absorbs light of a given frequency it absorbs one quantum only."

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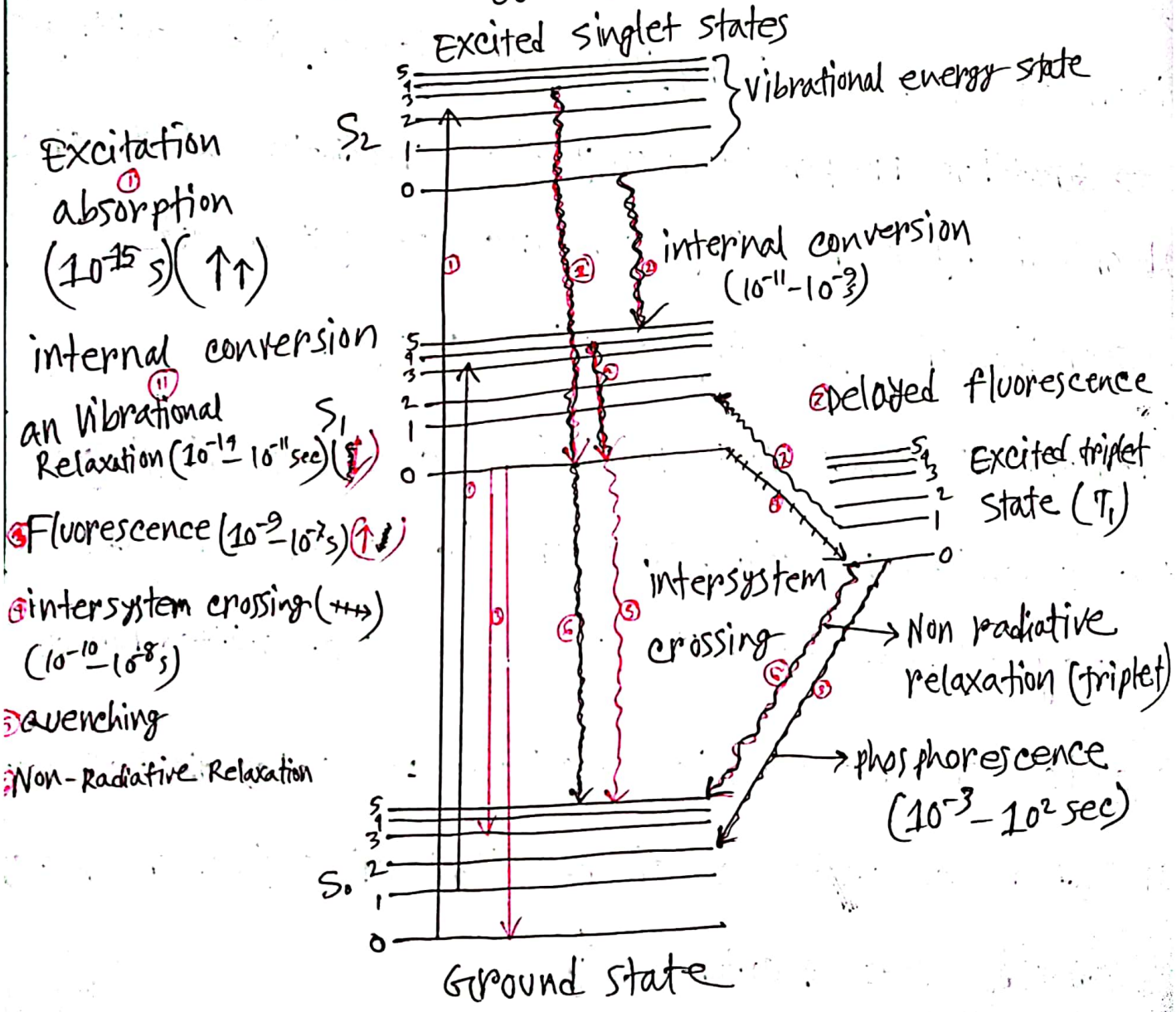
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⇒ Einstein and stark law : "Each molecule which take part in a chemical reaction absorb one quantum of light which induces the reaction."

For activation of one mole, $E = Nh\nu$.

(*) 2018, 2018 (U)
Draw Jablonski energy diagram & show photochemical physical processes with their time scale ?

⇒ Jablonski energy diagram



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2018, 2016, 2018 (U)

(*) What is quantum yield? Why does the value of quantum yield deviate from unity? Give examples where the quantum yield is greater than unity and less than unity.?

⇒ Q.Y এর Definition লেখা হয়েছে আগে।

⇒ The quantum yield is the ratio of the number of events that occur (such as the formation of a product) to the number of photons absorbed by a system. A value of unity (1) implies that every absorbed photon results in the desired event. The Q.Y often often deviates from unity due to several factors: (I) Non-Radiative Decay process (II) competing photophysical processes (III) photochemical side reactions (IV) Inadequate energy transfer (V) quenching effect (VI) thermal deactivation.

⇒ Example: greater than unity ($\Phi > 1$):

$CO + Cl_2 \rightarrow COCl_2$: $\Phi = 10^3$:

$H_2 + Cl_2 \longrightarrow 2HCl$ $\Phi = 10^4$ to 10^6

$2H_2O_2 \longrightarrow 2H_2O + O_2$ $\Phi = 7$

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less than unity ($\varphi < 1$).

$H_2 + Br_2 \longrightarrow 2HBr \quad \varphi = 0.01$

$2NH_3 \longrightarrow N_2 + 3H_2 \quad \varphi = 0.2$

$CH_3COCH_3 \longrightarrow CO + C_2H_6 \quad \varphi = 0.1$

2022(t)

Write down the application of gas, chemical and dye lasers?

⇒ gas lasers:

laser type	
Helium-Neon laser	spectroscopy, barcode scanning, alignment, optical demonstrations.
Argon laser	Refinal phototherapy
N_2 laser	pumping of dye laser, measuring air pollution, scientific research.
CO_2 laser	Material processing (welding), photocoustic spectroscopy.

dye layers: Research, laser medicine, spectroscopy, birth mark removal, isotope separation.

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chemical lasers:

laser type	Applications
HF laser	used in research for laser weaponry by the U.S.
oxygen-iodine laser	laser weaponry, scientific and material research.
All gas-phase iodine laser	scientific, weaponry, aerospace.

2016

⊗ A substance absorbs 2×10^{16} quanta of radiations per second and 0.002 mole of it reacts in 1209 sec, calculate quantum yield of the reaction?

$$\Rightarrow Q = \frac{\text{No. of molecules reacting in a given time}}{\text{No. of quanta absorbed in the same time}}$$

Now,

No. of quanta absorb in 1209 sec, $= 1209 \times 2 \times 10^{16}$

$= 2408 \times 10^{16}$

No of molecules reacting in 1209 sec $= 0.002 \times 6.023 \times 10^{23}$

$$\therefore Q = \frac{0.002 \times 6.023 \times 10^{23}}{2408 \times 10^{16}}$$

$= 50$ Ans.

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2018(t), 2018

The decomposition of HI vapour was carried out with $\lambda = 207$ nm radiation. Absorption of each calorie of energy gave 1.49×10^{-5} g of hydrogen. What is the Quantum yield?

$\Rightarrow$ No. of moles reacting $= \frac{1.49 \times 10^{-5}}{128} = 1.125 \times 10^{-7}$ mol

No. of Einestein absorbed $= Nh\nu = Nh \times \frac{c}{\lambda}$

$= 6.023 \times 10^{23} \times 6.626 \times 10^{-34} \times \frac{3 \times 10^8}{207 \times 10^{-9}}$

$= 578382.58\ J = 137710.14$ cal

$\therefore Q = \frac{1.125 \times 10^{-7}}{137710.14}$ mol cal^{-1} $= 8.17 \times 10^{-13}$ mol cal^{-1}

Ans.

2020, 2015, 2022(t)

The dipole moment (μ) & polarisability (α) of Chlorobenzene are 1.57D and 1.23×10^{-23} cc & dens at 25°C is 1.1171 g cc^{-1}. Calculate dielectric constant of the chlorobenzene?

$\Rightarrow$ we Know,

$$\frac{\epsilon - 1}{\epsilon + 2} \cdot \frac{M}{\rho} = \frac{4}{3}\pi N\alpha + \frac{4}{3}\pi N \frac{\mu^2}{3KT}$$

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$$\Rightarrow \frac{\epsilon-1}{\epsilon+2} = \frac{4\pi N\rho}{3M}\left(\alpha + \frac{\mu^2}{3KT}\right)$$

$$\Rightarrow \frac{\epsilon-1}{\epsilon+2} = \frac{4\times 3.1416\times 6.023\times 10^{23}\times 1.1171}{3\times 112.56}\left\{1.23\times 10^{-23} + \frac{(1.57\times 10^{-18})^2}{3\times 1.38\times 10^{-16}\times 298}\right\}$$

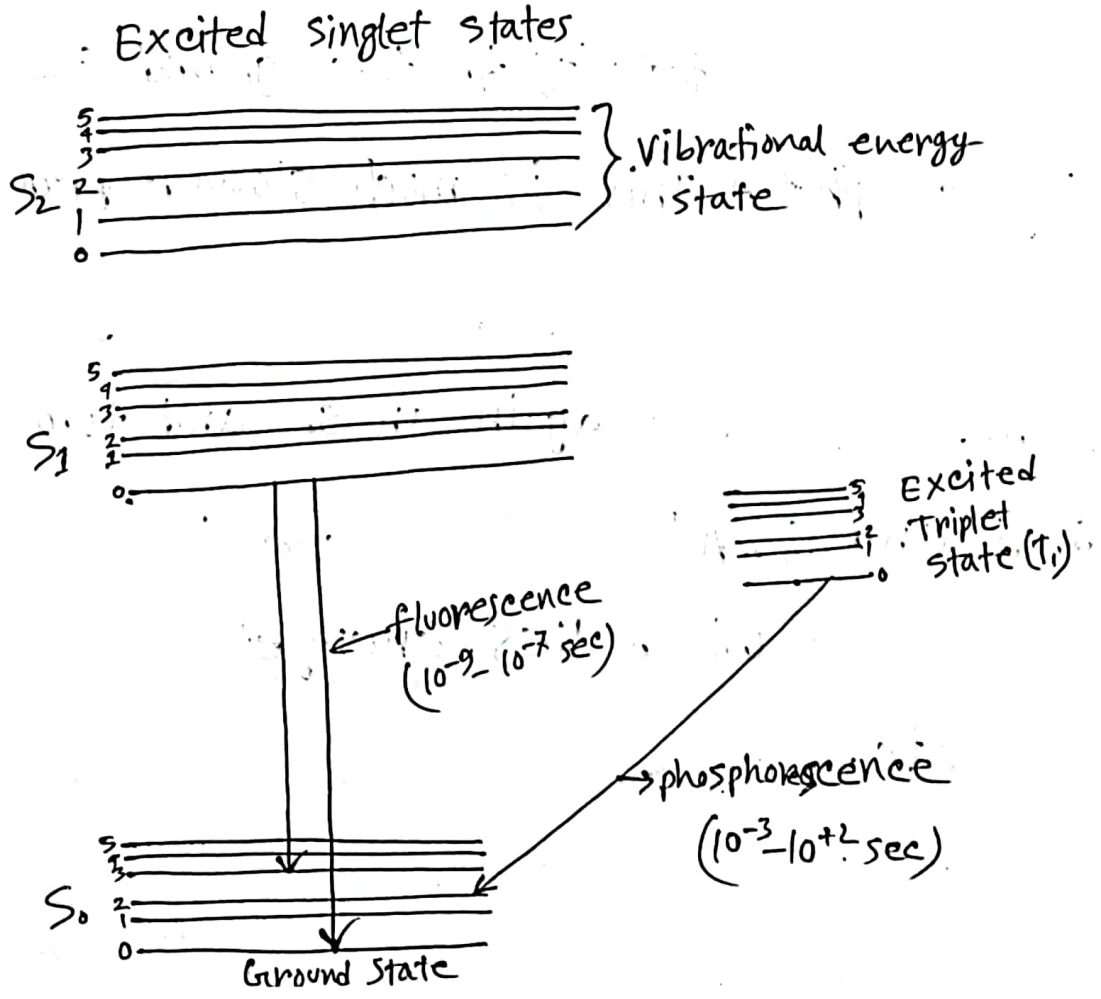
$$\Rightarrow \frac{\epsilon-1}{\epsilon+2} = 0.80823 \Rightarrow \epsilon - 1 = 0.80823\epsilon + 1.62$$

$$\therefore \epsilon = 13.64 \quad \text{Ans.}$$

2021

Explain the terms fluorescence and phosphorescence using Jablonski diagram?

⇒



In this diagram Ground state (S_0), Excited singlet state (S_1, S_2...) Excited Triplet states (T_1, T_2...).

Fluorescence occurs when an e^- in an excited sing State (usually S_1) returns to the ground state (S_0), emitti a photon in the process, due to some energy loss in the relaxation process. This process is typically a very fast process and occurs on 10^{-9} to 10^{-7} sec.

phosphorescence involves a longer-lived excited state, usually a triplet state (T_1). The e^- eventually transitions from the triple state (T_1) to the ground state (S_0), emitting a photon. This is much slower process, occur in 10^{-3} to 10^2. For phosphorescence, things are a bi different. There are again an S_0 and two excited states, S_1 and S_2. Additionally there is an excited triplet T_1 state which lies energetically btwn the S_0 and S_1 state.

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(★) 2017, 2019(t) What is photosensitization? Discuss how the mercury atom acts as a photosensitizer?

⇒ Photosensitization is the phenomenon in which light insensitive reaction can be made ligh sensitive by adding a small amount of foreign material which can absorb light and stimulate the reaction without taking part in the reaction. Such material known as photosensitizer.

⇒ Dissociation of H_2 into atom at 253.7 nm in the presence of Hg vapour.

$$Hg(g) + H_2 \longrightarrow 2H + Hg(g) \quad \text{at } 253.7 \text{ nm}$$

Hg atom absorbs light $\longrightarrow$ Hg* collide with H_2 $\longrightarrow$ dissociate into atoms.

Mechanism:

$$Hg + h\nu \longrightarrow Hg^*$$

$$Hg^* + H_2 \longrightarrow H_2^* + Hg$$

$$H_2^* \longrightarrow 2H$$

(★) 2017, 2019(t) What are the primary and secondary process? Briefly discuss the consequences of light absorption by an atom and a molecule?

⇒ Process in which light radiation is absorbed by an atom or

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molecule giving rise to the formation of an excited atom or an excited molecule is known as primary process.

$$A + h\nu \longrightarrow A^*$$

processes which involves the excited atoms, molecules or free radicals produced in the primary process is known as secondary processes.

⇒ consequences of light absorption by an atom:

(I) primary effect:

(a) electronic excitation: one or more e^- go into higher energy levels.

(b) Ejection of e^-: If excitation is high, one or more e^- ma given out by atom; causes ionization. Average lifetime of electronically excited atom is 10^{-7} to 10^{-8} s.

(II) secondary effect

(a) Fluorescence: excited e^- returns to ground state, emitti a part or whole of excitation energy.

(b) Resonance fluorescence: excited e^- return to ground state emitting radiation of exactly same frequency of incident radiation.

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(c) phosphorescence: In some cases emission of radiation persist for some time after the incident light is cut off.

(d) Photochemical reaction: excited atoms may take part in chemical reactions with non-excited atoms or molecules

(e) photosensitization: an excited atom may collide with a molecule and by energy exchanges may cause the molecules to dissociate.

consequences of light absorption by a molecule:

(I) Primary:

(a) Excitation of molecules: includes electronic, vibrational & rotational excitation.

(b) Isomeric change to form new molecules.

(c) Dissociation of molecules. (d) Ionisation of molecules through ejection: known as photoionisation.

(II) Secondary:

(a) photosensitization reaction (b) delayed photochemical changes.

(c) Rise in T. (d) Fluorescence (e) phosphorescence.

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2021(t)

Define: fluorescence, resonance fluorescence and phosphorescence?

⇒ fluorescence: it is the emission of light by a substance that has absorbed electromagnetic radiation.

When a beam of light is incident on certain substance they emit visible light or radiations and they stop emitting light or radiation as soon as the incident light is cut off. This phenomenon is known as fluorescence. It is a form of luminescence.

resonance f.! In certain cases, the fluorescent light has the same frequency as that of ~~light~~ incident light. This fluorescent light is called resonance radiation & the phenomenon is known as resonance fluorescent. This phenomenon is analogous to resonance of sound. For example when Hg vapour with atoms in normal state ($^1S_0$) is exposed to ultraviolet radiations, it was found to absorb the radiation of $\lambda = 2537$ Å. It then converts to $^3P_1$. If excited e⁻ return to the ground state, they emit the radiations of the same λ, i.e. 2537 Å.

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$Hg(^1S_0) + h\nu \longrightarrow Hg^*(^3P_1)$

$Hg^*(^3P_1) \longrightarrow Hg(^1S_0) + h\nu$

phosphorescence: When a beam of light is incident on certain substances they emit light continuously even after the incident light is cut off. This type of delayed fluorescence ~~are~~ is called phosphorescence. The life time of phosphorescence (10^{-4} to 20 sec) is much longer than fluorescence (10^{-6} to 10^{-4} sec).

(A) 2021(t) Radiation of λ 2540Å was passed through a cell containing 10mL a Sln of 0.0495M oxalic acid and a 0.01M uranyl sulphate. After the absorption of 8.81×10^8 ergs of radiation the conc. of oxalic acid was reduced to 0.0383M. calculate the Q.Y. ?

10ml of 0.0495 M oxalic acid = 0.000495 mole

10ml of 0.0383 M oxalic acid = 0.000383 mole

Amount of oxalic acid decomposed = (0.000495 − 0.000383)

= 0.000112 mole

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No. of molecules $= 0.000112 \times 6.023 \times 10^{23}$

$= 6.7457 \times 10^{19}$

Number of of quanta absorbed $= \frac{8.81 \times 10^8 \text{ ergs}}{h\nu}$

$= \frac{8.81 \times 10^7 \text{ J} \times \lambda}{hc} = \frac{8.81 \times 10^7 \times 2540 \times 10^{-10}}{6.626 \times 10^{-34} \times 3 \times 10^8}$

$= 1.1267 \times 10^{20}$

So, $Q.Y = \frac{6.7457 \times 10^{19}}{1.1267 \times 10^{20}} = 0.5987$

Ans